Sorting Machine



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Sorting machines

What it is used for

A sorting machine separates agricultural products such as seeds, grains, fruits, or vegetables by size, weight, colour, or other specified characteristics. Use suitable settings for the crop and inspect the output to maintain product quality and reduce damage.

Main parts

- Feed hopper or loading conveyor
- Sorting screens, rollers, belts, or channels
- Drive motor and transmission
- Sensors or grading controls, if fitted
- Collection bins and discharge outlets
- Protective guards and control panel

Before you start

1. Confirm that the machine is suitable for the crop and sorting task.
2. Inspect the hopper, screens, belts, rollers, guards, and drive components.
3. Check that the emergency stop and other safety devices function correctly.
4. Remove stones, metal, and other foreign objects from the incoming material.
5. Ensure collection bins are clean, correctly positioned, and labelled if required.
6. Wear suitable PPE and clear the work area of unnecessary people and obstacles.

How to use

1. Place the machine on a stable surface and secure it as instructed.
2. Fit the required screen, guide, or sorting attachment while the machine is isolated.
3. Set sorting parameters according to the crop and required output grade.
4. Position collection bins under the correct discharge outlets.
5. Close all guards and confirm that feed and discharge paths are clear.
6. Switch on the machine and run it empty briefly to check normal operation.
7. Feed material gradually at the recommended rate.
8. Monitor sorting accuracy, material flow, vibration, and unusual noise.
9. Inspect samples from each outlet and adjust settings using the approved procedure if necessary.
10. Stop feeding and allow remaining material to clear before shutdown.
11. Switch off and isolate power before removing jams or cleaning internal parts.
12. Collect and label sorted products according to the workplace procedure.
13. Clean the machine and record output, settings, and any faults.

⚠ Safety

- Only trained operators should operate or adjust the sorting machine.
- Keep hands and tools away from moving belts, rollers, screens, and drive parts.
- Never bypass guards, interlocks, or emergency stops.
- Isolate power and verify complete stoppage before clearing jams or servicing.
- Control dust and use suitable respiratory protection where required.
- Use safe lifting methods when handling heavy crop containers.

Care & storage

- Remove crop residue and dust after each production run.
- Inspect screens, belts, bearings, fasteners, and guards regularly.
- Keep sensors and sorting surfaces clean and unobstructed.
- Lubricate only specified points according to the maintenance schedule.
- Report damaged components, inaccurate sorting, or abnormal vibration.

Common mistakes to avoid

- Feeding material too quickly and causing blockages or poor sorting.
- Using an incorrect screen or unsuitable grading setting.
- Ignoring worn screens, dirty sensors, or loose fasteners.
- Clearing a blockage while the machine is energised.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.